

GREATER NOIDA INSTITUTE OF TECHNOLOGY

(Engineering Institute)

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FORMAT FOR B.TECH PROJECT SYNOPSIS

YOJNA MITRA AI

Your AI Guide to Government Schemes

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Discovery Platform |

1. INTRODUCTION

In recent years, the expansion of digital governance in India has led to the introduction of a vast array of Central and State government schemes spanning critical socio-economic sectors such as education, employment, agriculture, healthcare, housing, women and child welfare, scholarships, financial assistance, entrepreneurship, and general social welfare. While administrative bodies continually launch supportive initiatives, citizens frequently encounter severe friction in discovering schemes tailored to their specific demographic and financial profiles. The primary bottleneck is not a scarcity of welfare programs, but rather the complexity of navigation—citizens often lack awareness regarding their eligibility criteria, mandatory supporting documentation, application channels, and jurisdictional boundaries (Central versus State-specific).

Although centralized portals like *myScheme* offer personalized scheme discovery through structured filters and questionnaires, ordinary users—particularly those less comfortable with complex digital interfaces—often struggle with keyword searches, exhaustive drop-down menus, and formal bureaucratic terminology. **Yojna Mitra AI** addresses this digital divide by introducing an intuitive, AI-powered conversational agent that bridges the gap between citizens and government portals through natural-language interaction.

Field of the Project

This project lies at the intersection of Artificial Intelligence, Natural Language Processing (NLP), Conversational Systems, and E-Governance Technology. It incorporates human-computer interaction principles, intelligent profile extraction, automated rule-based eligibility matching engines, and responsive web deployment frameworks.

Technologies Used

- **Frontend:** Streamlit (Python-based framework for fast, interactive chat-driven user interfaces without complex boilerplate code).
- **Backend & AI Processing:** Python, utilizing Large Language Models (LLMs) and custom Natural Language Processing (NLP) pipelines for intent recognition and demographic profile extraction.
- **Eligibility Matching Engine:** Deterministic rule-based matching system implemented in Python to evaluate normalized user attributes against structured government scheme datasets.

- **Data Management:** Structured JSON / Relational database schemas maintaining verified scheme metadata (age limits, income thresholds, occupational codes, benefit summaries, and official application URLs).

Key Features of the System

- **Natural Language Interaction:** Empowers users to state their personal circumstances in colloquial language rather than filling rigid web forms.
- **Automated Profile Extraction & Normalization:** Translates conversational inputs (e.g., colloquial income descriptions, regional abbreviations) into standardized database attributes.
- **Relevance Ranking:** Evaluates multi-variable eligibility rules and presents schemes categorized by match strength (High, Medium, Possible).
- **Transparent Explanation Layer:** Clearly articulates the rationale behind each recommendation (e.g., matching student status, state residency, and annual income brackets).
- **Actionable Guidance:** Provides comprehensive benefit details, mandatory document checklists, and direct official government URLs.

Technical Terms

- **Conversational Agent / Chatbot:** An AI program designed to simulate human conversation via text or voice interactions, capable of multi-turn dialogue management.
- **Profile Extraction:** The computational process of identifying and isolating key demographic, financial, and occupational entities from unstructured natural language text.
- **Data Normalization:** The process of organizing data attributes into standard, uniform formats (e.g., converting text-based income expressions into numeric values) to ensure reliable comparison.
- **Rule-Based Eligibility Engine:** A deterministic computational module that evaluates conditional logic (IF-THEN statements) to verify whether a user profile satisfies official statutory criteria.
- **Minimum Viable Product (MVP):** A version of a new product with just enough features to be usable by early customers who can then provide feedback for future product development.

2. LITERATURE REVIEW

The integration of artificial intelligence and natural language processing into public service delivery and citizen-facing e-governance has gained substantial momentum globally. Academic research and real-world implementations provide key insights into conversational information retrieval and automated rule-matching systems.

Prior Research and Academic Contributions

1. **Conversational Agents in E-Governance:** Research by Sharma et al. (2021) examined the efficacy of conversational interfaces in bridging the digital literacy gap among rural and semi-urban citizens accessing public welfare portals. Their study demonstrated that natural-language query models significantly reduced task completion time and user abandonment rates compared to traditional multi-page form architectures.
2. **Automated Rule-Based Matching Systems:** A study by Kumar and Rao (2020) investigated hybrid frameworks combining large language models for unstructured text understanding with deterministic rule engines for high-stakes decision verification. Their findings emphasized that while LLMs excel at intent parsing and conversational dialogue, financial and statutory eligibility decisions must rely on hard-coded deterministic rules to guarantee absolute accuracy and prevent algorithmic hallucination.
3. **Semantic Normalization in Public Sector Databases:** Work by Patel et al. (2022) focused on semantic mapping frameworks designed to translate colloquial user expressions into structured ontological attributes. Their experiments proved that preprocessing user inputs through data normalisation layers drastically improves retrieval precision in recommendation engines handling government schemes and scholarships.

Existing Applications and Tools

Existing platforms such as *myScheme* (National Portal of India) provide centralized discovery, eligibility checking, and repository features for hundreds of welfare programs. While robust, *myScheme* relies heavily on structured filtering parameters, category selections, and questionnaire forms. Commercial customer support chatbots deployed across various state portals offer basic FAQ retrieval but lack deep semantic profile extraction and multi-turn eligibility reasoning.

Table 1: Comparison of Existing Portals vs. Proposed Yojna Mitra AI

Platform / Tool	Interaction Model	Profile Extraction	Conversational Reasoning
myScheme Portal	Filter & Form-based	Manual form entry	Limited / Questionnaire-driven
Standard Gov Portals	Keyword Search	None	Static FAQs only
Yojna Mitra AI (Proposed)	Natural Language Chat	Automated NLP Extraction	Multi-turn Conversational Guidance

While existing platforms successfully catalog government schemes, there remains a distinct operational gap in delivering a frictionless, conversational discovery experience for users unfamiliar with bureaucratic taxonomies. Yojna Mitra AI addresses this gap by combining LLM-driven conversational comprehension with strict, deterministic eligibility matching.

3. OBJECTIVES

The primary objective of this project is to design and develop **Yojna Mitra AI**, an intelligent, conversational government-scheme discovery platform that simplifies access to welfare programs through natural-language interaction.

Main Objective

To construct a hybrid system combining Natural Language Processing (for user profile extraction and conversational dialogue) with a deterministic rule-based eligibility engine, enabling citizens to discover, understand, and apply for relevant government schemes through natural dialogue.

Sub-Objectives

- **To design a conversational chat interface:** Developed using Streamlit to facilitate seamless, multi-turn dialogue without requiring complex web development boilerplate.
- **To implement an AI profile extraction module:** Capable of accurately parsing unstructured user statements to identify key demographic, financial, and occupational attributes.
- **To develop a data normalization pipeline:** To translate colloquial user inputs (e.g., regional state abbreviations, informal income figures) into standardized database variables.

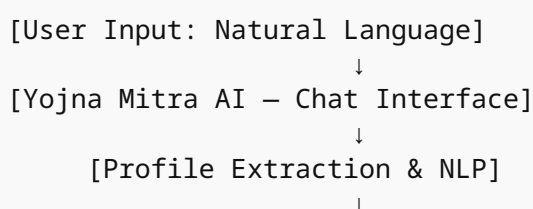
- **To construct a rule-based eligibility matching engine:** That evaluates normalized user profiles against structured scheme criteria (age, income, occupation, social category, state residency) to determine precise eligibility.
- **To build a relevance ranking and explanation layer:** That categorizes matching schemes by relevance (High, Medium, Possible) and provides clear, human-readable explanations of why a user qualifies.
- **To compile a verified scheme knowledge base:** Maintaining structured metadata, benefit breakdowns, required documentation lists, and official government application URLs.

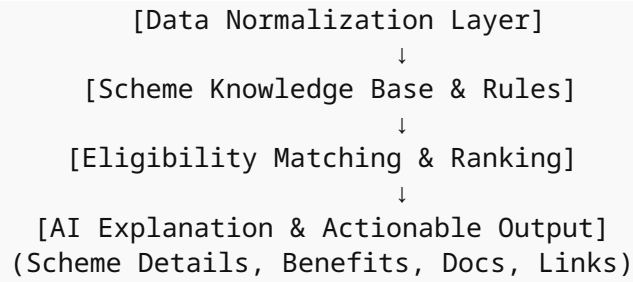
4. METHODOLOGY / PLANNING OF WORK

The project follows a modular, phased development lifecycle structured across five core functional modules assigned to the project team.

Step-by-Step Methodology

1. **Phase 1 — Data Collection & Structuring (Data Team):** Compiling verified scheme details from official portals (such as the National Portal of India), establishing JSON/database schemas covering age limits, income caps, benefits, and source URLs.
2. **Phase 2 — AI & NLP Pipeline (AI/NLP Module):** Designing prompt architectures and parsing logic to extract user attributes (age, state, income, occupation) from natural language inputs and managing multi-turn dialogue flow.
3. **Phase 3 — Eligibility & Ranking Engine (Eligibility Module):** Developing the logical matching rules (evaluating minimum age, maximum income caps, state jurisdiction, and occupational requirements) and sorting schemes by relevance.
4. **Phase 4 — Frontend Development (Frontend Module):** Building the user interface using Streamlit, featuring an interactive chat window, dynamic scheme result cards, and structured detail views.
5. **Phase 5 — Integration & Testing (Integration Module):** Combining all modules into a cohesive prototype, executing test cases with diverse user profiles, validating recommendation accuracy, and deploying the application.





5. REFERENCES

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